

Connor Savugot

B.S. in Computer Engineering | North Carolina State University | Graduated May 2026

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Embedded firmware engineer and **NCSU computer engineering** graduate. Four internships and full-time experience spanning embedded systems, electrical engineering, and manufacturing automation. Experienced in system-level troubleshooting across hardware, firmware, and software. Developed C/C++ firmware, test automation tools, and manufacturing databases.

SKILLS

Programming & Data: C, C++, SQL, VBA, Microsoft Access, Excel, Python, MATLAB, Bash

Architectures: STM32, dsPIC33, PIC32, TI AM62, TMS320, MSP430; bare-metal firmware, FreeRTOS, Embedded Linux

Peripherals: GPIO, DAC control, ADC sensing and filtering, PWM control, timers, interrupts, watchdogs, and communications

Interfaces: CAN 2.0, CAN FD, DroneCAN, J1939, I2C, IPMI, SPI, UART/RS-232, Ethernet, TCP/IP

Hardware & Manufacturing: Schematic interpretation; datasheet validation, PCB review in Altium, KiCad, and P-CAD; MCU programming/debug interfaces, oscilloscopes, logic/protocol analyzers; production programming, test automation, tooling data

EXPERIENCE

AEGIS Power Systems | Embedded Firmware Engineer

May 2026–Present

- ▷ Develop and debug C firmware; perform system testing using software logs, protocol analyzers, oscilloscopes, and multimeter measurements to validate hardware and software behavior.
- ▷ Isolate root causes and verify corrective actions by assessing system-level failures across PCBs, wiring, power, firmware, CAN/Ethernet, Linux services, and application software using schematics, datasheets, and diagnostic tools.
- ▷ Support firmware development, hardware integration, and documentation for VPX/IPMI power systems, TI AM62 Embedded Linux controllers; troubleshoot Yocto/Arago images, device trees, boot issues, CAN power devices, and watchdogs.

AEGIS Power Systems | Embedded Systems Intern | Computer Science Intern

Summers 2024 & 2025

- ▷ Wrote bare-metal C firmware for a TMS320-controlled bidirectional power converter with 12 PWM A/B pairs (24 outputs); configured frequency, period, duty cycle, and phase shift for each A output and set its paired B output to follow or invert it.
- ▷ Worked on a multi-processor power system where a FreeRTOS PIC32 managed communications and DSP firmware updates over TCP/IP, SPI, UART, and CAN/J1939, while a C2000 TI DSP executed real-time converter controls.
- ▷ Simplified TI's C++ CLI serial flasher for C2000 devices by removing unused device, CPU2, and dual-boot options, allowing production staff to program precompiled firmware with the click of a single button.
- ▷ Established Git-based development and release processes to maintain firmware-to-hardware traceability, preserve defect and revision history, support multi-engineer development, and provide production with controlled software releases and documentation.

TEAM Industries | Engineering Intern

Summers 2021 & 2022

- ▷ Built a configurable validation and duplicate-detection system for a 17,000+ record plant-wide tooling database, allowing users to select comparison criteria while automatically identifying, locating, and highlighting duplicate or inconsistent manufacturing records.
- ▷ Developed VBA-driven inspection tools that validated production measurements against engineering tolerances, enforced data-entry rules, and automatically classified inspected parts as ACCEPT, REJECT, or INCOMPLETE.

PROJECTS

EV Active Sensor Adapter | Battery Monitoring & CAN

NC State University

- ▷ Wrote STM32G474 firmware for a custom EV sensor board that read ten ADC thermistors, exchanged CRC-checked cell and fault data with a BQ79616 BMS Chip over UART, and transmitted measurements over CAN.

C++ Maze Solver

NC State University

- ▷ Developed an object-oriented C++ maze-solving application using reusable classes, dynamic data structures, and multiple search strategies; debugged and compared implementations for correctness, memory use, and execution performance.

C++ Microarchitecture Simulators

NC State University

- ▷ Developed modular C++ simulators for cpu processor subsystems using classes and objects to model system state, component interactions, and configurable behavior; validated results across multiple test configurations.

TEACHING EXPERIENCE

ECE 306 Embedded Systems | Teaching Assistant & Course Mentor

Spring 2025–Spring 2026

- ▷ Taught supplemental lessons and helped students debug MSP430FR2355 firmware and custom PCBs across three semesters, including ADC sensors, PWM motor control, timers, wiring, soldering, and toolchain problems.

IEEE LEADERSHIP

IEEE at NC State | OPS2 Lead & Parts Distribution Coordinator

Fall 2025–Spring 2026

- ▷ Led two semesters of OPS2, IEEE's ten-week applied electronics course that paired classroom theory with hands-on soldering, PCB and BOM development, embedded programming, PlatformIO, and GitHub.
- ▷ Built and managed an ECE 306 parts program that cut student costs and raised \$10,000 for IEEE in two weeks.